Additional Table 3 Characteristics of studies on trehalose intervention for animal models of synucleinopathy

First author (year) Country	Blinding of assessor	Intervention(s), Control and setting [Intervention schedule]	Animal characteristics [Neurodegenerative diseases]	Outcome measure(s)	Clinical outcome(s)	Physiological changes	Adverse event(s)	Conclusion
Darabi et al. (2019) [11] Iran	N/R	**Intervention:** Trehalose pre-treatment. [3%, oral administration, daily, 3 d] **Control:** None.	**6-OHDA rats [PD]** Intervention: *n* = 10 Control: *n* = 10 **WT** Control: *n* = 10 **Background** Wistar rats **Age (mon)** N/R **Gender** 30 Male	<u>Clinical</u> Number of rotations <u>Biochemical</u> Degree of dopaminergic deficit - DA - DOPAC Expression of antioxidant transcription factor - Nrf2 - GR - GPx - CAT Expression of autophagy marker - LC3II/LC3I ratio - p62	Trehalose pre-treatment improved motor behaviour and reduced apomorphine-induced rotations.	Trehalose pre-treatment attenuated 6-OHDA-induced reduction of DA and DOPAC. Trehalose pre-treatment induced autophagy flux (increased LC3II/LC3I ratio and p62). Trehalose increased expression of Nrf2 transcription factor and downstream antioxidant transcription (GR, GPx, and CAT).	N/R	Trehalose pre-treatment improved motor function in PD by protecting dopaminergic neuronal loss through activation of autophagy and reduction of oxidative stress.
Ferguson et al. (2015) [12] United States	N/R	**Intervention[1]:** Trehalose. [1%, oral administration, daily, 38 d] + MPTPp [25 mg/kg MPTP in	<u>Clinical</u> **MPTPp mice [PD]** Intervention[1]: *n* = 10 Control[1]: *n* = 10 **WT**	<u>Clinical</u> Locomotor activity Buried food test - Olfaction Motor coordination Grip strength - Neuromuscular function	Trehalose did not alleviate any of the clinical measures in MPTPp mice.	Trehalose attenuated MPTP-induced reduction of DA, DOPAC, HVA and 5-HIAA in striatum of MPTPp mice.	N/R	1% trehalose is insufficient to confer neuroprotection for behavioural alteration in PD.

	saline, 250 mg/kg probenecid in 0.1 M NaOH and buffered in Tris-HCl, injection, 2x/wk, 5 wk] **Intervention[2]:** Trehalose. [1%, oral administration, daily, 38 d] **Control[1]:** MPTPp [25 mg/kg MPTP in saline, 250 mg/kg probenecid in 0.1 M NaOH and buffered in Tris-HCl, 2x/wk, 5 wk] **Control[2]:** Saline [0.9%, injected into right lateral ventricles] **Control[3]:** Probenecid [250 mg/kg, in 0.1 M NaOH and buffered in Tris-HCl, 2x/wk, 5 wk]	Intervention[2]: *n* = 10 Control[2]: *n* = 10 **Background** C57Bl/6 **Age (mon)** 1.73 **Gender** 40 Male Biochemical **MPTPp mice [PD]** Intervention[1]: *n* = 5 Control[1]: *n* = 5 Control[3]: *n* = 5 **Background** C57Bl/6 **Age (mon)** 1.73 **Gender** 15 Male	Hole board exploration - Anxiety, stress, & neophilia Biochemical Degree of dopaminergic deficit - DA - DOPAC - 5-HIAA - HVA				
He et al. (2016) [13] China	Yes	**Intervention[1]:** Trehalose. [5%, oral administration, daily, 6 wk] **Intervention[2]:** Trehalose	**AAV α-Synuclein transgenic mice [PD]** Intervention[1]: *n* = 10 Intervention[2]: *n* = 10 Intervention[3]: *n* =	Clinical Cylinder test - Locomotor asymmetry Biochemical Degree of dopaminergic	2% & 5% Trehalose improved and maintained locomotor asymmetry in transgenic mice.	2% & 5% Trehalose protected DA neurons in SN and TH in striatum of transgenic mice. 2% & 5% Trehalose cleared or prevented	N/R	Trehalose improved motor function in PD by protecting against α-synuclein-mediated neurodegeneration through activation of autophagy.

	(2%, oral administration, daily, 6 wk) **Intervention[3]:** Trehalose (0.5%, oral administration, daily, 6 wk) **Control:** Glucose (5%, oral administration, daily, 6 wk)	10 Control: *n* = 10 **WT** *n* = 10 **Background:** Sprague-Dawley **Age (mon)** 2 **Gender** 60 Female	deficit - DA neurons - TH - DA - DOPAC - HVA - DA turnover (HVA/DA) Level of plagues - α-Synuclein Expression of autophagy marker - LC3II	α-synuclein accumulation in SN and striatum of transgenic mice. 0.5%, 2%, & 5% Trehalose alleviated decreased concentration of DA and DA turnover in striatum of transgenic mice. 5% Trehalose alleviated decreased concentration of DOPAC in striatum of transgenic mice. 2% & 5% Trehalose increased LC3II in striatum of transgenic mice.

Howson et al. (2019) [14] Canada	Yes	**Intervention[1]:** Trehalose (2.67g/kg, oral administration, daily, 6 wk) **Intervention[2]:** Trehalose (0.89 g/kg, oral administration, daily, 6 wk) **Intervention[3]:** Trehalose (2%, oral administration, daily, 6 wk) **Intervention[3]:** Trehalose (2%, oral administration, daily, 6 wk) **Control:** Vehicle. (Sterile water)	Rodent **AAV α-Synuclein transgenic rats [PD]** Intervention[1]: *n* = 10 Intervention[2]: *n* = 10 Intervention[3]: *n* = 10 Control: *n* = 10 **WT** *n* = 10 **Background** Sprague-Dawley **Age (mon)** N/R **Gender** 50 Female Mammal **AAV α-Synuclein transgenic macaque [PD]** Intervention[1]: *n* = 8-9 Control: *n* = 8-9 **WT** *n* = 8-9 **Background** Cynomolgus **Age (mon)** N/R **Gender**	**Rodent** Clinical Cylinder test - Locomotor asymmetry Biochemical Degree of dopaminergic deficit - TH^+ cells - DA - DOPAC - HVA - DA turnover (HVA/DA) Level of plagues - α-Synuclein **Mammal** Clinical Motor activity Biochemical Degree of dopaminergic deficit - TH^+ cells - DA - DAT - DOPAC - HVA - DA turnover (HVA/DA) Level of plagues - α-Synuclein	Rodent 2.67g/kg daily trehalose improved locomotor asymmetry in transgenic rats. Mammal *Motor activity did not differ between transgenic and WT macaques.	Rodent 2.67g/kg daily trehalose has higher DA in striatum in transgenic rats. Mammal 2.67g/kg daily trehalose increased putamen DA and DAT in transgenic macaques.	N/R	Trehalose improved motor function in PD through improving dopaminergic deficits in striatum.

			25 Female				
Kakoty et al. (2021) [15] India	N/R	**Intervention[1]:** Trehalose. [4g/kg, oral administration, 1 mon] **Intervention[2]:** Sodium butyrate. [300mg/kg, oral administration, 1 mon] **Intervention[3]:** Trehalose & sodium butyrate. [2g/kg trehalose, 300mg/kg sodium butyrate, oral	**PFF α-Syn rats [PD]** Intervention[1]: $n = 8$ Intervention[2]: $n = 8$ Intervention[3]: $n = 8$ Control: $n = 8$ **Sham-control** Control: $n = 8$ **WT** Control: $n = 8$ **Background** Wistar **Age (mon)** 2-2.5	<u>Clinical</u> Rota-rod test - Motor coordination Open field test - Locomotion and anxiety Narrow beam walk - Gait Novel object location and recognition test - Spatial memory and learning <u>Biochemical</u> Degree of	Combination of trehalose and sodium butyrate improved motor and cognitive performances in the transgenic mice.	Combination of trehalose and sodium butyrate and sodium butyrate alone reduced the TNF-α, IL-1, IL-6, α-Synuclein, CRP, $GFAP^{+}$, and IBA1 in the substantia nigra of transgenic mice. Combination of trehalose and sodium butyrate and sodium butyrate alone increased TH^{+}, DA, DAT, MAP2, H3 acetylation, LC3II,	Combination of trehalose and sodium butyrate ameliorated motor and cognitive performances in PD through improving dopaminergic deficit via upregulating the expression autophagy flux and growth factor, and inhibition of reactive gliosis.

	administration, 1 mon] **Control:** None	**Gender** 48 Male	dopaminergic deficit - TH^+ - DA - DAT - MAP2 Level of plagues - α-Synuclein Expression of inflamattory cytokine - TNF-α - IL-1 - IL-6 - CRP Expression of autophagy marker - H3 acetylation - Lamp2 - LC3II/β-actin ratio - SQSTM1/β-actin ratio Expression of growth factor - BDNF Expression of astrogliosis - $GFAP^+$ Expression of microgliosis - IBA1		SQSTM1, BDNF, and H3 acetylation in transgenic mice.		
Lin et al. (2020) [16] Taiwan	N/R	**Intervention[1]:** Trehalose (2%, oral administration, 1.5 mon) **Intervention[2]:** Lactulose (2%, oral administration, 1.5	**MPTP mice [PD]** Intervention[1]: $n = 8$ Intervention[2]: $n = 8$ Intervention[3]: $n = 8$ Control: $n = 8$ **WT** Control: $n = 8$	<u>Clinical</u> Pole test - Bradykinesia Stride length - Gait <u>Biochemical</u> Degree of dopaminergic deficit	Trehalose, lactulose and melibiose improved motor activity and gait in transgenic mice.	Trehalose, lactulose and melibiose did not ameliorate reduction of TH^+ neurons in transgenic mice. Trehalose improved striatal DA greater than lactulose and melibiose in	N/R	Trehalose, lactulose, and melibiose improved motor function in PD through improving dopaminergic deficits via upregulating the expression of antioxidant transcription factors,

	mon) **Intervention[3]:** Melibiose (2%, oral administration, 1.5 mon] **Control:** None	**Background** C57BL6J **Age (mon)** 2 **Gender** 40 Male	- TH^+ cells - DA - DAT Expression of antioxidant transcription factor - Nrf2 - SOD2 - NQO1 Degree of oxidative stress - 4-HNE Expression of autophagy marker - LC3II/LC3I ratio Expression of astrogliosis - $GFAP^+$ Expression of microgliosis - IBA1		transgenic mice. Trehalose, lactulose and melibiose upregulated Nrf2, NQO1, LC3II/LC3I ratio in transgenic mice. Trehalose, lactulose and melibiose inhibited the upregulation of 4-HNE, $GFAP^+$ and IBA1 in transgenic mice. Lactulose and melibiose ameliorated reduction of DAT and SOD2 in the ventral midbrain of transgenic mice.		autophagy flux, and inhibition of reactive gliosis.
Pupyshev et al. (2019) [17] Russia	**Intervention[1]:** Trehalose. [2%, oral administration, daily, 7 d] + Rapamycin [10mg/kg, oral administration, daily, 7 d] + MPTP [20mg/kg, injection, daily, 4 d] **Intervention[2]:** Trehalose [2%, oral administration,	**MPTP mice [PD]** Intervention[1]: *n* = 5-7 Intervention[2]: *n* = 5-7 Control[1]: *n* = 5-7 Control[2]: *n* = 5-7 Control[3]: *n* = 5-7 **WT** Control[4]: *n* = 5-7 **Background** C57BL6J **Age (mon)** 3	<u>Clinical</u> Passive avoidance test - Memory & learning Open field test - Locomotion and anxiety <u>Biochemical</u> Expression of autophagy marker - LC3II Degree of dopaminergic deficit - TH	Trehalose, rapamycin, and combination improved memory and learning in MPTP mice; no additive effect was observed. MPTP mice have increased locomotion compared to WT; no effect of rapamycin and trehalose were observed.	Trehalose increased LC3-II in striatum of MPTP mice, which is suppressed by co-administration of rapamycin. Trehalose and co-administration with rapamycin increased LC3-II in SN of MPTP mice. Trehalose and rapamycin restored TH in SN of MPTP mice; greater effect was observed with co-administration.	N/R	Trehalose improved cognitive and motor function in PD through improving dopaminergic deficits in nigrostriatal system by activating autophagy. Additive effect was observed in co-administration with rapamycin in biochemical but not clinical performances.

	daily, 7 d] + MPTP [20mg/kg, injection, daily, 4 d] **Control[1]:** Rapamycin [10mg/kg oral administration, daily, 7 d] + MPTP [20mg/kg, injection, daily, 4 d] **Control[2]:** Solvent of rapamycin [Ethanol diluted 20-fold with 5% Tween 80 containing 5% PEG-400, oral administration, daily, 7 d] + MPTP [20mg/kg, injection, daily, 4 d] **Control[3]:** MPTP [20mg/kg, injection, daily, 4 d] **Control[4]:** Saline [injection, daily, 4 d]	**Gender** Male			Trehalose and co-administration with rapamycin restored TH in striatum of MPTP mice.		
Pupyshev et al. (2021) [18]	N/R	**Intervention[1]:** Trehalose (2%, oral	**MPTP mice [PD]** Intervention[1]: $n \geq 5$	<u>Clinical</u> Conditioned passive avoidance response	Trehalose, rapamycin, and combination	Trehalose and co-administration with rapamycin exerted no	N/R	An enhanced therapeutic effect of the combined

Russia	administration, 2 wk) **Intervention[2]:** Rapamycin. (2 mg, intraperitoneal injection, 2 wk) **Intervention[3]:** Trehalose + Rapamycin [2%, oral administration, 2 wk (Trehalose) + 2 mg, intraperitoneal injection, 2 wk (Rapamycin)] **Intervention[4]:** Trehalose + Rapamycin + 3-methyladenine [2%, oral administration, 2 wk (Trehalose) + 2mg, oral administration, 2 wk (Rapamycin); 15 mg/kg intraperitoneal injection, 2 wk (3-methyladenine)] **Control[1]:** Rapamycin solvent. (2 mg, oral administration, 2 wk) **Control[2]:** None.	Intervention[2]: $n \geq 5$ Intervention[3]: $n \geq 5$ Control[1]: $n \geq 5$ Control[2]: $n \geq 5$ **WT** Control[2]: $n \geq 5$ **Background** C57BL6 **Age (mon)** N/R **Gender** N/R **B6.Cg-Tg mice [early PD]** Intervention[1]: $n \geq 5$ Intervention[2]: $n \geq 5$ Intervention[3]: $n \geq 5$ Intervention[4]: $n \geq 5$ Control[1]: $n \geq 5$ Control[2]: $n \geq 5$ **Background** A53T **Age (mon)** 5 **Gender** N/R	- Learning and memory <u>Biochemical</u> Expression of autophagy marker - LC3II Degree of dopaminergic deficit - TH	improved memory and learning in MPTP mice; no additive effect was observed.	change in the LC3-II in SN and striatum of MPTP and early PD mice. Trehalose and co-administration with rapamycin (compared to trehalose alone) increased TH in the SN and striatum of MPTP mice, with greater effect in early PD mice.	treatment was revealed on the expression of TH especially in the early PD stage but not sufficient to translate into cognitive performance.

Sarkar et al. (2014) [19] United States	N/R	**Intervention[1]:** Trehalose (2%, oral administration, daily, 38 d) + MPTPp (25 mg/kg MPTP in saline, 250 mg/kg probenecid, injection, 2x/wk, 5 wk) **Control[1]:** MPTPp [25 mg/kg MPTP in saline, 250 mg/kg probenecid in 0.1 M NaOH and buffered in Tris-HCl, 4x/wk, 5 wk] **Control[2]:** Probenecid (250 mg/kg, 2x/wk, 5 wk) **Control[3]:** Saline	**MPTPp mice [PD]** Intervention[1]: *n* = 16 Control[1]: *n* = 16 **WT** Control[2]: *n* = 16 Control[3]: *n* = 16 **Background** C57BL6 **Age (mon)** 1.87 **Gender** 64 Male	<u>Clinical</u> N/R <u>Biochemical</u> Degree of dopaminergic deficit - TH - DAT - DA - DOPAC - 5-HIAA - HVA Expression of astrogliosis - $GFAP^+$ Expression of microgliosis - CD11b Integrity of endothelial cells Expression of glucose transporter - GLUT-1 Integrity of tight junction proteins - Occludin - ZO-1	N/R	*No difference in probenecid and control-treated WT. Trehalose pre-treatment prevented MPTP-induced reduction of DAT, and TH in striatum and SN of MPTPp mice. Trehalose pre-treatment prevented MPTP-induced reduction of DA, 5-HIAA and HVA in striatum of MPTPp mice. Trehalose pre-treatment prevented MPTP-activated $GFAP^+$ in striatum and SN of MPTPp mice. Trehalose pre-treatment prevented MPTP-activated CD11b in striatum and SN of MPTPp mice. Trehalose pre-treatment prevented MPTP-induced endothelial cell clusters and sclerosis in striatum and SN of MPTPp mice. Trehalose pre-	N/R	Trehalose confers neuroprotection against damage to micro-vessels and endothelial in PD pathology.

					treatment prevented MPTP-induced reduction of GLUT-1 in striatum of MPTPp mice. Trehalose pre-treatment prevented MPTP-induced reduction of occluding and ZO-1 in striatum of MPTPp mice.

Tanji et al. (2015) [20] Japan	N/R	**Intervention:** Trehalose (2%, oral administration, daily, 12 wk) **Control[1]:** Maltose (2%, oral administration, daily, 12 wk) **Control[2]:** Sucrose (2%, oral administration, daily, 12 wk) **Control[3]:** Vehicle (normal drinking water)	**α-Synuclein A53T transgenic mice [LBD]** Intervention: *n* = N/R Control[1]: *n* = N/R Control[2]: *n* = N/R Control[3]: *n* = N/R **WT** Intervention: *n* = N/R Control[1]: *n* = N/R Control[2]: *n* = N/R Control[3]: *n* = N/R **Background** C57BL/6J **Age (mon)** 2.8 – 7.5 **Gender** 55 N/R	Clinical N/R Biochemical Expression of autophagy - LC3II - Phospho-Beclin-1/Total beclin-1 Level of plagues - α-Synuclein Expression of chaperones - HSP90 - SigmaR1	N/R	Expressions of LC3II was higher in trehalose than maltose-treated transgenic mice. Expressions of HSP90 and SigmaR1 was higher in trehalose than maltose-treated transgenic mice. Trehalose decreased level of α-Synuclein in transgenic mice. Trehalose increased phospho-beclin-1 to total beclin-1 ratio in transgenic mice.	N/R	Trehalose decreased level of protein aggregates in LBD through activation of autophagy and chaperone for protein folding.

[+]: Positive; 4-HNE: 4-hydroxynonenal; 5-HIAA: 5-hydroxyindoleacetic acid; 6-OHDA: 6-hydroxydopamine; CAT: catalase enzyme; DA: dopamine; DAT: dopaminergic transporter; DOPAC: 3,4-dihydroxyphenylacetic acid; GFAP: glial fibrillary acidic protein; GLUT-1: glucose transporter 1; GPx: glutathione peroxidase; GR: glutathione reductase; HSP90: heat shock protein 90; HVA: homovanilic acid; IBA1: ionized calcium-binding adaptor molecule 1; LBD: Lewy body disease; LC3: microtubule-associated protein 1A/1B-light chain 3; MPTP: 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine; N/R: not reported; NQO1: NAD(P)H dehydrogenase; Nrf2: nuclear factor (erythroid-derived 2)-like 2; PD: Parkinson's disease; SigmaR1: Sigma-1 receptor; SN: substantia nigra; SOD2: superoxide dismutase2; TH: tyroxine-hydroxylase; WT: wild-type; ZO-1: Zona-occludens-1.